

Karollyn Larissa de Quadros

Environmental Engineer, M.Sc. Candidate

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Research Interests

Hydrology & Water Resources: Hydrological Balance; Evapotranspiration Estimation; Terrestrial Water Storage; Basin-Scale Hydrology; Low-Flow Rainfall-Runoff Modeling (GR Models).

Remote Sensing & Data Analytics: Satellite Gravimetry (GRACE/GRACE-FO); Hydroclimatic Time Series Analysis; Bias Correction Methods; Machine Learning Applications.

Modeling & Natural Hazards: Hydrodynamic Modeling (Delft3D); Reservoir Hydrodynamics; Landslide Early Warning Systems; Highway Slope Risk Assessment; Extreme Hydroclimatic Events.

Education

M.Sc. in Environmental Engineering (In Progress) Aug 2024 – Present
Graduate Program in Environmental Engineering (PPGEA), Federal University of Paraná (UFPR), Brazil

Qualification Exam: Scheduled for December 15, 2025

B.Sc. in Environmental Engineering Jan 2018 – Aug 2024
Federal University of Paraná (UFPR), Brazil

Honors & Academic Awards

- **UFPR Gold Medal (Academic Distinction):** First place ranking in the Environmental Engineering graduating class (2024).
- **1st Place / Winner of the SIMEPAR Corporate Hackathon (2026):** Designed and prototyped an innovative solution for real-time monitoring and landslide risk assessment along highway slopes.
- **Maximum Grade (10/10) in Capstone Project:** Awarded for final graduation project on reservoir hydrodynamics (2024).
- **UFPR Scientific Initiation Award:** Distinction for undergraduate research on internal wave dynamics in reservoirs (2023).
- **SBRH Conference Presentation Distinction:** Recognized research work presented at the XXV Brazilian Symposium on Water Resources (2023).

Academic & Research Experience

Graduate Researcher (M.Sc.) – PPGEA / UFPR Aug 2024 – Present

- Conducting research on basin-scale evapotranspiration estimation across Paraná state using water balance equations.
- Integrating in situ hydrometeorological station datasets with downscaled GRACE/GRACE-FO terrestrial water storage data.
- Analyzing space-time hydroclimatic variability and validating remote sensing data against ground-truth stations.

Undergraduate Researcher (Scientific Initiation) – Hydrology Lab, UFPR 2022 – 2024
Advisor: Prof. Dr. Tobias Bleninger

- Developed numerical hydrodynamic models (Delft3D) to evaluate the impact of floating solar panels on lake internal wave dynamics.
- Configured numerical grids and executed thermal stratification simulations in lakes and reservoirs.

Undergraduate Researcher (Scientific Initiation) – Meteorology Lab, UFPR 2019 – 2021
Advisor: Prof. Dr. Alice Marlene Grimm

- Performed statistical analysis of hydroclimatic time series to assess climate variability impacts on South American precipitation patterns.

Professional & Applied Research Experience

M.Sc. Researcher & Scholarship Holder – SIMEPAR Aug 2024 – Present

- Applied remote sensing (GRACE/GRACE-FO) and meteorological models to operational hydrology and drought management.
- Developed a pilot project for landslide early warning systems using precipitation critical threshold curves in Serra do Mar.
- Supported the development of a regional drought status index for public water supply utilities.

Research Intern – SIMEPAR Sep 2023 – Aug 2024

- Executed statistical programming in R for operational environmental datasets and remote sensing data validation.

Environmental Engineering Intern – Lactec Apr 2021 – Apr 2022

- Processed, validated, and assured quality control (QA/QC) for environmental and hydrological datasets.

Environmental Engineering Intern – RHA Engenharia e Consultoria Apr 2020 – Apr 2021

- Assisted in environmental impact studies, data consistency checks, and technical documentation.

Publications & Manuscripts

Peer-reviewed Conference Papers

- **de Quadros, K. L.**, Ferreira, E. M., Fernandes, C. V. S., Bleninger, T. B., & Bueno, R. C. (2023). *Influence of Floating Photovoltaic Panels on Internal Wave Dynamics in Lakes and Reservoirs*. XXV Brazilian Symposium on Water Resources (SBRH).
- **de Quadros, K. L.**, Lima, M. F. D. S., Ferreira, D. M., Gonçalves, J. E., Mannich, M., Inouye, R. T., & Silveira, R. (2025). *Bias Correction of Daily Precipitation Data from the WRF Model*. XXVI Brazilian Symposium on Water Resources (SBRH).

Manuscripts in Preparation

- **de Quadros, K. L.**, et al. *Estimation of Evapotranspiration in River Basins of Southern Brazil Using Hydrological Balance and GRACE/GRACE-FO Data*.
- **de Quadros, K. L.**, et al. *Rainfall Accumulation Thresholds for Landslide Early Warning Systems: A Case Study from the Serra do Mar, Brazil*.

Technical Skills & Methods

- **Programming Languages:** R, Python.
- **Software & Modeling:** Delft3D, QGIS, L^AT_EX, MS Office Suite, Linux OS.
- **Data Analytics & Formats:** Spatial/Gridded Data Processing (NetCDF, HDF5, GeoTIFF, Shapefiles), Automated Data Extraction (APIs), Data Cleansing & QA/QC, Hydroclimatic Time Series Analysis.
- **Methodologies:** GRACE/GRACE-FO Remote Sensing, Bias Correction Methods, Extreme Value Analysis, Conceptual Rainfall-Runoff Modeling (GR Models), Statistical Regressions, Machine Learning Fundamentals.

Languages

Portuguese: Native | **English:** Advanced (Reading, Writing, Speaking) | **French:** Intermediate